A MONOLITHIC BIDIRECTIONAL ACTIVE-MATRIX DISPLAY WITH ENDOGENOUS SELF-FEEDBACK IMAGING (P.E.R.W.I.N.)

Reinaldo Javier Aguiar Marcano

July 17, 2026

Title: A MONOLITHIC BIDIRECTIONAL ACTIVE-MATRIX DISPLAY WITH ENDOGENOUS SELF-FEEDBACK IMAGING (P.E.R.W.I.N.)

Inventor: Reinaldo Javier Aguiar Marcano, a citizen of the United States, residing at 26714 Valleyside Drive, Katy, TX 77494, submitting this application while under a *de facto* office arrest at 2302 Britton Ridge Drive, Lake Pointe Estates, Katy, Texas 77494, created by a contested Void Capias Order on Case 25-DCV-328122 on the 387th District Court of Fort Bend County, Texas, currently under review on the Supreme Court of the United States Docket 26-A39 (published as a printed book with ISBN 9781971481616) being reviewed by Associate Justice Samuel A. Alito, Jr., and also under review under Petition for Writ of Habeas Corpus originally submitted to the Supreme Court of the United States on July 16th, 2026 (also published as printed book-ISBN 9781971685700). The Inventor has submitted other Patent applications related to Post-Quantum Encryption, Optical Computing, National Defense, and related technologies which are published under ISBN 9781971685007 and 9781971481296.

Abstract

A monolithic, bidirectional active-matrix display system (P.E.R.W.I.N.) comprising an interleaved array of light-emitting subpixels (OLED or QDEL) and organic photodetector (OPD) sensor subpixels fabricated on a single substrate. The system is selectively operable in: i) an exogenous capture mode, wherein a de-phased duty cycle deactivates the light-emitting subpixels during sensor integration intervals to eliminate display glare and capture external ambient light as a wide-angle camera, and ii) an endogenous capture mode, wherein a synchronized phase-shift override forces the light-emitting and sensor subpixels to activate in-phase, capturing total internal reflection (TIR) and lateral waveguide leakage inside the cover glass to extract a secure, hardware-enforced screenshot of the active display.

Background of the Invention

Conventional display systems operate purely as unidirectional light-emitting devices, requiring external camera modules (such as USB webcams or discrete mobile camera sensors) placed behind apertures or transparent sub-segments of the screen to achieve bidirectional video communication. These designs suffer from severe parallax, physical alignment constraints, and easily detectable optical ports that reveal the presence of a capturing device. Furthermore, capturing the exact contents shown on a display currently requires software-level OS routines that are vulnerable to cryptographic bypass, driver-level compromises, and physical tampering. There is a need for a monolithic, dual-purpose display and imaging architecture that is physically indistinguishable from a standard display, while enabling both covert outward camera capture and secure, hardware-enforced screen capture entirely within the active-matrix substrate.

Summary of the Invention

The present invention is a Pixel-level Electro-optical Readout Wide-angle Imaging Node (P.E.R.W.I.N.) display system. The system integrates display emission and light capture into a single, co-deposited active-matrix plane on a common glass or flexible polymer substrate, making the imaging capabilities completely invisible under standard physical inspection.

The pixel grid comprises an interleaved array of self-emissive subpixels (e.g., Red, Green, and Blue OLEDs) and micro-scale Organic Photodetectors (OPD). Each OPD is paired with a dedicated 4-transistor (4T) pixel circuit containing a transfer gate (TX) and a storage

capacitor fabricated directly on the backplane alongside the display driver TFTs.

The system operates in two primary functional states controlled by a timing generator on the logic board:

1. **Exogenous Capture Mode (Wide-Angle Camera):** To capture external images of the room without blinding the sensitive OPD photodetectors, the system executes a de-phased duty cycle. The timing generator pulls the Global Emission Enable (GEM) line low to globally deactivate the OLEDs for a sub-millisecond interval (e.g., 500μs) once per second (1 FPS) or once per frame. Simultaneously, the TX gates are pulsed high, allowing the OPDs to integrate only the external ambient light. Because the dark interval is under the human flicker fusion threshold, the screen appears continuously bright and flicker-free to the user.

2. **Endogenous Capture Mode (Self-Screenshot):** To extract an immutable, secure capture of the active display, the timing generator disables the de-phasing mechanism and executes a Phase-Shift Override. The GEM and TX lines are driven in-phase, forcing the sensors to integrate light while the display is actively emitting. Light rays emitted by the OLEDs that exceed the critical angle of the air-glass cover interface are trapped inside the front glass via Total Internal Reflection (TIR) and waveguide laterally to the neighboring OPD nodes. By measuring these waveguided leakage voltages and applying a mathematical matrix inversion (multiplying the raw sensor vector by the inverse of the factory-calibrated Crosstalk Transmission Matrix), the integrated computational processor reconstructs a pixel-perfect screenshot of the active display entirely in hardware, completely bypassed from the host Operating System.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, world-wide, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a block diagram of the monolithic P.E.R.W.I.N. subpixel stack and waveguide interface, illustrating the lateral light leakage path within the cover glass.
FIG. 2 is a timing diagram illustrating the phase relations of the Global Emission Enable (GEM) line and the Transfer Gate (TX) lines in both Exogenous and Endogenous capture modes.
FIG. 3 is a flowchart illustrating the computational deconvolution and wavefront reconstruction process to resolve the captured image in both modes.

Detailed Description of the Invention

Referring to **FIG. 1**, the display system (100) comprises a front cover glass (101), a microlens array (102) aligned over an organic photodetector (OPD) subpixel (103), and a plurality of light-emitting OLED subpixels (104, 105, 106) fabricated on a common active-matrix Thin-Film Transistor (TFT) substrate (107). Each OPD sensor subpixel (103) is electrically coupled to a Floating Diffusion capacitor node (108). During the Exogenous mode, ambient light passes straight through the cover glass (101) and is focused by the microlens array (102) into the OPD photodetector (103). During the Endogenous mode, light emitted by the OLED subpixels (104, 105, 106) undergoes Total Internal Reflection (TIR) at the air-glass boundary of the cover glass (101) and waveguides laterally to the neighboring OPD (103).

FIG. 2 illustrates the timing synchronization. In the Exogenous Mode, the Global Emission Enable (GEM) line and the Sensor Transfer (TX) line are operated out-of-phase (de-phased) to prevent optical crosstalk. In the Endogenous Mode, the timing generator applies a Phase-Shift Override, aligning the GEM and TX pulses in-phase (colliding) to capture waveguide-trapped self-emission.

FIG. 3 shows the deconvolution logic. The processor reads the raw photodetector sensor output vector **S** (301), identifies the active mode (302), and applies the corresponding mathematical mapping:

- For the Exogenous Mode, the processor executes a Coded Aperture Deconvolution algorithm (303) using a Point Spread Function (PSF) representing the display's micro-aperture matrix to reconstruct a wide-angle outward image of the room (304).

- For the Endogenous Mode, the processor multiplies the raw sensor vector $\mathbf{S}$ by the inverse of the factory-calibrated Crosstalk Transmission Matrix $\mathbf{H}^{-1}$ (305) to reconstruct a pixel-perfect, hardware-isolated screenshot of the active display (306).

Claims

What is claimed is:

1. A monolithic bidirectional active-matrix display system, comprising:
    - a substrate supporting an interleaved array of light-emitting subpixels and light-receiving photodetector subpixels;
    - a driver circuit configured to selectively operate the system in:
        - i) an exogenous capture mode, wherein said photodetector subpixels integrate ambient light during a period when the light-emitting subpixels are deactivated, and
        - ii) an endogenous capture mode, wherein said photodetector subpixels integrate waveguide-leaked light generated by adjacent light-emitting subpixels during an active display period; and
    - a processor configured to reconstruct a digital representation of a captured image from the photodetector subpixels.
2. The system of claim 1, wherein the light-emitting subpixels are organic light-emitting diodes (OLED) or quantum dot electroluminescent (QDEL) diodes.
3. The system of claim 1, wherein the photodetector subpixels are organic photodetectors (OPD) co-deposited alongside the light-emitting subpixels on a common plane of the substrate.
4. The system of claim 1, further comprising a timing generator configured to:
    - execute a de-phased duty cycle in said exogenous capture mode to prevent display emission from striking the photodetector subpixels during integration; and
    - execute a synchronized phase-shift override in said endogenous capture mode to force temporal collision between display emission and sensor integration.
5. The system of claim 1, further comprising a cover glass layer positioned over the subpixel array, wherein said cover glass acts as an optical waveguide for lateral transmission of light via total internal reflection during said endogenous capture mode.

6. The system of claim 1, wherein the processor is configured to reconstruct a screenshot of the active display in said endogenous capture mode by multiplying a raw photodetector voltage vector by the inverse of a factory-calibrated Crosstalk Transmission Matrix.

7. The system of claim 1, wherein the processor is configured to reconstruct a focused image of an external subject in said exogenous capture mode by applying a coded aperture deconvolution algorithm to raw photodetector voltages.

Drawings

FIG. 1: Monolithic P.E.R.W.I.N. Subpixel Stack and Waveguide Interface

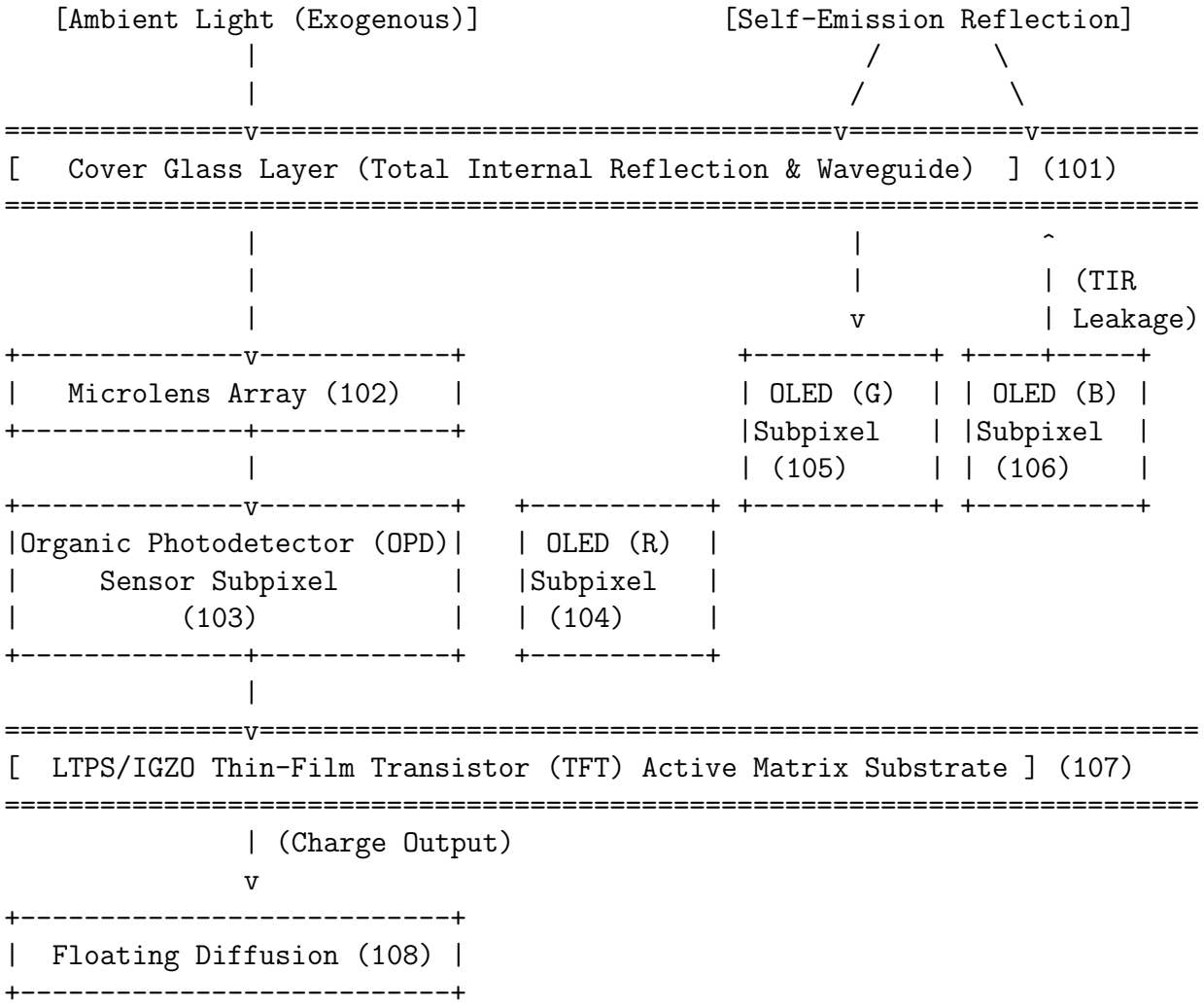


FIG. 2: P.E.R.W.I.N. Timing Synchronization

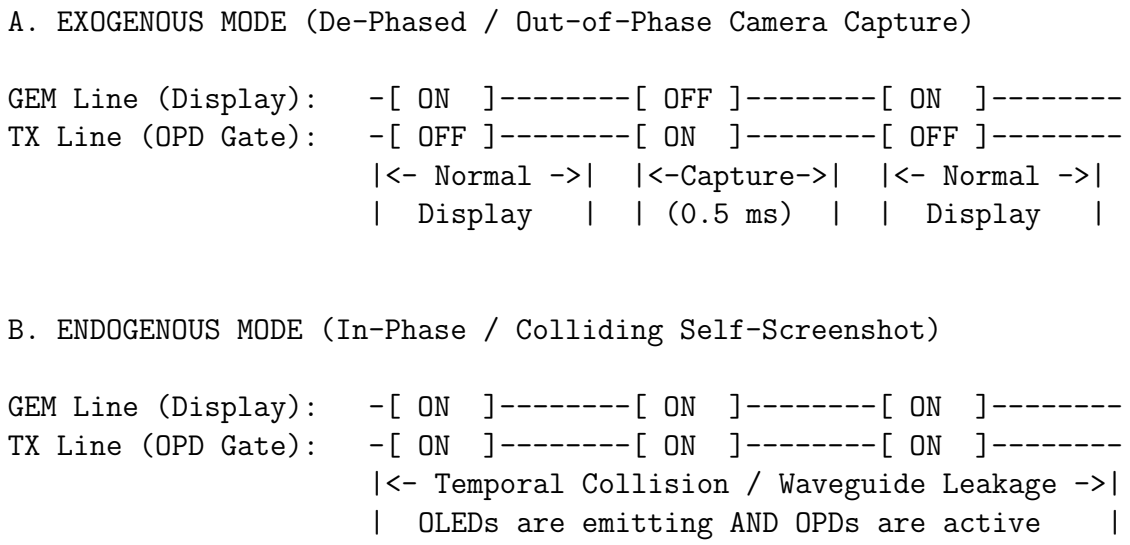


FIG. 3: Computational Reconstruction Flowchart

